Speculative Decoding: How It Evolved, When It Stays Lossless, and What's Next

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Abstract

Every LLM you use generates one token at a time. This is autoregressive decoding, a major bottleneck of inference: producing n tokens takes n passes through a model with tens of billions of parameters. Speculative decoding accelerates this with a draft-and-verify loop, and it now runs under nearly every hosted LLM: OpenAI, Anthropic, DeepSeek, and Kimi all ship it in production. This education material teaches the concept through three questions: How It Evolved (how does speculative decoding work, and how did four generations of draft models build upon each other?), When It Stays Lossless (does the guarantee hold beyond math and code?), and What's Next (what are the exciting new directions?). Learners walk through three generations of draft models (EAGLE-3, DFlash, DSpark), then a hands-on lab on a single GPU: serve an accelerated model, measure acceptance and speed, and adjust the acceptance threshold in deployment. Prepared materials: (1) interactive website[1], (2) LosslessBench[2], (3) lab code[3].

1 Concept Summary

The material follows the progression a learner needs, each section building on the previous one:

- *Why it matters:* speculative decoding runs under nearly every hosted LLM, so the speed and quality of decoding affect everyone. OpenAI credited its 80% GPT-5.6 Luna price cut partly to a redesigned draft model (OpenAI, 2026), Anthropic's fast mode serves the same Claude Opus model up to 2.5× faster (Anthropic, 2026), DeepSeek ships DSpark for a 51% throughput gain, and Kimi K3 ships with its own draft model (Kimi Team, 2026).
- *How it works:* a lightweight draft model proposes the next γ tokens, the target model verifies all of them in a single forward pass at nearly the cost of one, and rejection sampling guarantees the output follows the target distribution (Leviathan et al., 2023; Chen et al., 2023).
- *Three generations, each removing one bottleneck:* **EAGLE-3** raises acceptance length with feature-level drafting (Li et al., 2025), **DFlash** cuts drafting time by generating the whole block in one diffusion pass (Chen et al., 2026), **DSpark** cuts verification time with confidence-scheduled trimming (DeepSeek, 2026), and DFlash 2 pushes acceptance length further (Inco AI, 2026). An interactive walkthrough carries one worked example across all 4 decoding algorithms.
- *What's next:* the material closes with the new directions: multimodal speculative decoding, speculating tool calls instead of tokens, and frontier labs absorbing acceleration into the model itself.

To answer whether speculative decoding methods generalize across domains beyond coding and math, which account for only 17% of real traffic (OpenRouter, 2026), we built LosslessBench, a five-domain evaluation, and a hands-on lab on a single GPU: learners serve Qwen3-8B with the three released draft checkpoints, reproduce published acceptance lengths, race the decoding methods across domains, and adjust the acceptance threshold themselves. As the threshold relaxes from 1.0 to 0.4, acceptance length climbs 48% while the generated pages drift, which learners judge with their own eyes.

[1] `https://neurips2026-speculative-decoding.vercel.app/`
[2] `https://huggingface.co/datasets/lilyzhng/lossless_bench`
[3] `https://github.com/lilyzhng/2026_NeurIPS_Education`

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Education Track.

2 Prerequisites and Technical Level

Level: introductory-to-intermediate; accessible to any engineer or student who knows that an LLM generates text one token at a time. **Prerequisites:** basic familiarity with transformer inference. The rejection-sampling losslessness argument is presented intuitively; hands-on labs are optional for participants who want more depth.

3 Learning Objectives and Outcomes

After engaging with this resource, a learner can:

- Understand why autoregressive decoding is the major bottleneck of LLM inference.
- Recognize how widely speculative decoding runs in production today (OpenAI, Anthropic, DeepSeek, Kimi), and why decoding speed and quality affect everyone.
- Explain why speculative decoding is fast, and why it is lossless in theory.
- Trace the evolution of draft models from the 2023 origin through EAGLE-3, DFlash, DSpark, and DFlash 2, and name the bottleneck each generation removes.
- Evaluate a speculative decoding model across domains with LosslessBench.
- Serve and evaluate speculative decoding models hands-on, including adjusting the acceptance threshold in deployment.

4 Grounding in NeurIPS Research (2022–2026)

The resource covers recent papers at NeurIPS and its sister venues:

- **Speculative decoding** (ICML 2023): the foundational draft-and-verify algorithm and its losslessness proof, the subject of Section 1.
- **EAGLE-3** (NeurIPS 2025): feature-level drafting with multi-layer fusion and a training-time test, the current autoregressive-draft baseline, Section 1.1.
- **DFlash** (ICML 2026): a block-diffusion drafter that generates the whole block in one forward pass, over $6\times$ lossless acceleration and up to $2.5\times$ over EAGLE-3, Section 1.2.
- **DSpark** (DeepSeek, 2026): confidence-scheduled verification on the block-parallel backbone, serving DeepSeek-V4-Flash 60–85% faster in production, Section 1.3.
- **ViSpec** (NeurIPS 2025): vision-aware speculative decoding, grounding the multimodal direction, Section 3.1.

Across venues over the years: blockwise parallel decoding (NeurIPS 2018), lossless speculative decoding (ICML 2023), EAGLE-3 (NeurIPS 2025), DFlash (ICML 2026); on the training side, FP8 in DeepSeek-V3, MXFP4 QAT in gpt-oss, and Kimi K3 shipping the speculator in post-training.

5 Teaching Materials Summary

All materials, including LosslessBench, are original and created for this track: (1) an **interactive self-contained website** following the How It Evolved / When It Stays Lossless / What's Next arc, with an interactive walkthrough and an acceptance-rate demo; (2) a **hands-on lab** with a Jupyter notebook walkthrough, every measured result embedded: serve Qwen3-8B with the three released draft checkpoints (EAGLE-3 vs DFlash vs DSpark), reproduce published acceptance lengths, race the decoding methods across domains, and measure per-domain performance on LosslessBench; (3) a **glossary** of key terms as a student handout, also served as tooltips throughout the website.

References

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- Chen et al. *Accelerating Large Language Model Decoding with Speculative Sampling.* 2023. `https://arxiv.org/abs/2302.01318`
- Li, Wei, Zhang, Zhang. *EAGLE-3: Scaling up Inference Acceleration of LLMs via Training-Time Test.* NeurIPS 2025. `https://arxiv.org/abs/2503.01840`
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- Zhang. *LosslessBench.* 2026. lilyzh.ng/writing/losslessbench